Tutorial - Week 10

TG 11 - TG 12

1. Determine whether the following improper integrals converge and find their value when possible.

(a) $\displaystyle\int_3^\infty \frac{1}{3x-1}\,dx$

(b) $\displaystyle\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$

(c) $\displaystyle\int_{-\infty}^0 \frac{2x+3}{(x^2+3x+1)^3}\,dx$

(d) $\displaystyle\int_0^{\frac{\pi}{2}} \sec^2 x\,dx$

(e) $\displaystyle\int_0^1 \frac{\sin\left(\frac{1}{x}\right)}{x^2}\,dx$

(f) $\displaystyle\int_0^3 \left(\frac{1}{\sqrt{9-x^2}}+\frac{1}{\sqrt[3]{3-x}}\right)dx$

(g) $\displaystyle\int_0^\infty \ln(x^2+2x+2)\,dx$

(h) $\displaystyle\int_{-\infty}^\infty \frac{1}{4x^2+4x+5}\,dx$

2. Which of the following improper integrals are convergent and which are divergent?

(a) $\displaystyle\int_\pi^\infty \frac{1}{x^2+\sin^2 x}\,dx$

(b) $\displaystyle\int_1^\infty \frac{1}{\sqrt{x^4+x}}\,dx$

(c) $\displaystyle\int_2^\infty \frac{1}{x\sqrt{x-1}}\,dx$

(d) $\displaystyle\int_1^\infty \frac{\pi+|\sin(x^3-1)|}{\sqrt{x}}\,dx$

(e) $\displaystyle\int_e^\infty \frac{1}{x(\ln x)^p}\,dx,\ p\in\mathbb{R}$

(f) $\displaystyle\int_0^\infty xe^{-2x}\,dx$

(g) $\displaystyle\int_{-\infty}^\infty \frac{1}{\cosh x}\,dx$

(h) $\displaystyle\int_0^{\frac{\pi}{2}} \csc x\,dx$

3. Determine the convergence or divergence of all the improper integrals defined by the following functions:

(a) $f(x)=\dfrac{1}{x^2-x}$

(b) $f(x)=\dfrac{1}{x^3-x^2}$

(c) $f(x)=\dfrac{1}{\sqrt{x}-x}$

(d) $f(x)=\tan x$

(e) $f(x)=\dfrac{1}{e^x-1}$

(f) $f(x)=\dfrac{1}{\sinh x}$

4. Use integration by parts to prove the convergence of the following improper integrals

(a) $\displaystyle\int_0^\infty \frac{\sin(x)}{x}\,dx$

(b) $\displaystyle\int_0^\infty \cos(x^2)\,dx$

5. Let $f\colon (0,+\infty)\to\mathbb{R}$ be defined by $f(x)=\dfrac{\ln x}{x^2+1}$.

(a) Use a suitable change of variable to find a relation between the improper integrals $\int_0^1 f(x)\,dx$ and $\int_1^\infty f(x)\,dx$.

(b) Prove that they are both convergent.

(c) Find $\int_0^{+\infty} f(x)\,dx$.

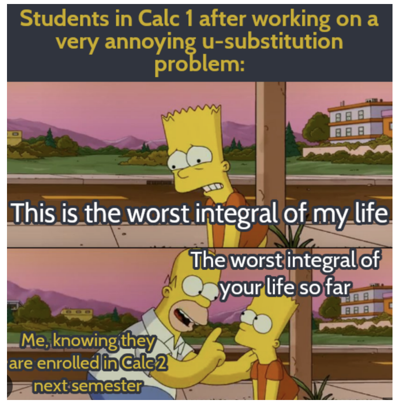
6. Find the length of the curve.

(a) $y=e^x$ from $x=0$ to $x=1$.

(b) $36y^2-(x^2-4)^3=0$ with $2\le x\le 3$ and $y\ge 0$.

(c) $y=\ln(\cos x)$ from $x=0$ to $x=\pi/3$.

(d) $\sqrt[3]{x^2}+\sqrt[3]{y^2}=c$ for any $c>0$.

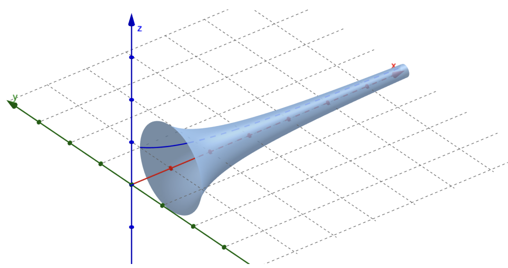


7. Find the area of the surface obtained by rotating the curve about the x-axis.

(a) $y = e^{-x}$ with $x \in [0, 1]$.

(b) $x^2 + y^2 = a^2$ with $x \in [0, \frac{a}{2}]$ and $y \geq 0$.

8. *Gabriel's Horn, Torricelli's Trumpet and the Painter's Paradox* For $t > 1$, define $A(t)$ as the area of the surface given by the revolution of the curve $y = \frac{1}{x}$ around the x-axis, between $x = 1$ and $x = t$. Also, define $V(t)$ as the volume enclosed by that surface. Show that $\lim\limits_{x\to\infty} A(t) = \infty$ but $\lim\limits_{x\to\infty} V(t)$ is finite.



9. Find the area of the surface obtained by rotating the curve about the y-axis.

(a) $x^2 + y^2 = a^2$ with $y \in [0, \frac{a}{2}]$ and $x \geq 0$.

(b) $\sqrt[3]{x^2} + \sqrt[3]{y^2} = 1$ with $0 \leq y \leq 1$.

(c) $(x - R)^2 + y^2 = r^2$, $0 < r < R$, from $y = -r$ to $y = r$.